

Abstention vs. Hallucination: Benchmarking LLM Source Attribution for Scientific Citations

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Abstract—Large language models (LLMs) increasingly generate citation-backed responses, yet citation hallucination remains a major challenge for trustworthy scientific information access. We introduce REASONS, a benchmark of 12,723 sentence-level citation instances spanning 12 arXiv subject categories, designed to evaluate scientific citation attribution under varying evidence conditions. We propose a dual-metric framework consisting of Abstention Rate (AR) and Hallucination Rate (HR) to characterize the trade-off between reliability and responsiveness. Using author-attribution and title-attribution tasks, we evaluate proprietary and open-source LLMs under zero-context, metadata-augmented, cascaded metadata-augmented prompting (CMP), retrieval-augmented, and adversarial settings. Advanced RAG lowers HR relative to Naïve RAG (65.4% vs. 87.6%) but reduces AR from 5.0% to 0%. Under adversarial metadata, several systems exceed 85% HR, while retrieval-augmented variants frequently maintain near-zero abstention. Human evaluation of 1,000 outputs ($\kappa = 0.78$) finds a 12.7:1 ratio of factual hallucinations to acceptable paraphrases. Our findings demonstrate that citation attribution systems should be evaluated not only for correctness but also for their ability to abstain appropriately under uncertainty. REASONS provides a benchmark and evaluation framework for studying attribution reliability in citation generation.

Index Terms—Scientific Citations, Attribution Evaluation, Retrieval-Augmented Generation, Large Language Models, Hallucination, Epistemic Safety

I. INTRODUCTION

Large language models (LLMs) increasingly generate citations to support their claims, yet citation hallucination poses a growing challenge to trustworthy scientific information access. Prior studies have documented fabricated and erroneous bibliographic citations in ChatGPT-generated responses [1], while more than 100 suspected citation hallucinations have recently been reported in NeurIPS 2025 accepted papers [2]. Recent theoretical work further argues that hallucination cannot be completely eliminated under general learning-theoretic conditions [3]. Existing work has investigated mechanisms and evaluation strategies for LLM source attribution and citation generation [4, 5]. However, a critical behavioral question remains underexplored: *when generating citations through parametric knowledge or retrieval, do systems abstain when evidence is insufficient, or do they confidently produce incorrect attributions?* As citation-aware systems like Perplexity [6]

and Bing [7] increasingly mediate scientific discovery, understanding attribution behavior has become essential. Figure 1 previews the overview of the REASONS benchmark construction and evaluation framework.

Prior work optimizes citation accuracy via better retrieval [8, 9] or training [10, 11], but ignores *when and how often systems should abstain rather than hallucinate*. Current benchmarks fail in three ways: paragraph-level evaluation masks claim-level errors [12, 13], Wikipedia datasets suffer LLM contamination [14, 15], and metrics ignore epistemic safety [16]. Critically, existing work cannot answer: *how frequently do systems abstain vs. hallucinate under varying evidence conditions, and does adding context shift behavior from cautious refusal to confident error?* Scientific citations offer sentence-level attributions with verifiable ground truth, minimal contamination, and domain variation, enabling controlled measurement of abstention-hallucination tradeoffs as evidence availability changes. We make the following contributions:

- 1) We introduce REASONS, the first sentence-level citation benchmark designed for behavioral evaluation, comprising 12,723 citations across 12 arXiv subject categories that enable systematic measurement of abstention and hallucination rates—metrics absent from existing attribution benchmarks.
- 2) We propose a dual-metric evaluation framework that jointly measures *abstention rate* (AR) and *hallucination rate* (HR) to characterize epistemic safety in citation generation. We demonstrate that standard attribution improvements create an abstention-hallucination tradeoff where increased responsiveness eliminates epistemic caution.
- 3) We validate our automated metrics through stratified human evaluation of 1,000 model outputs across five systems and multiple domains, demonstrating strong correlation ($r = 0.89$, $p < 0.001$) between automated HR and human-judged attribution errors. Our analysis confirms that hallucination is predominantly a factual error problem (12.7:1 ratio of factual errors to acceptable paraphrases) rather than a paraphrase detection issue, and reveals that systems struggle to calibrate abstention decisions appropriately.

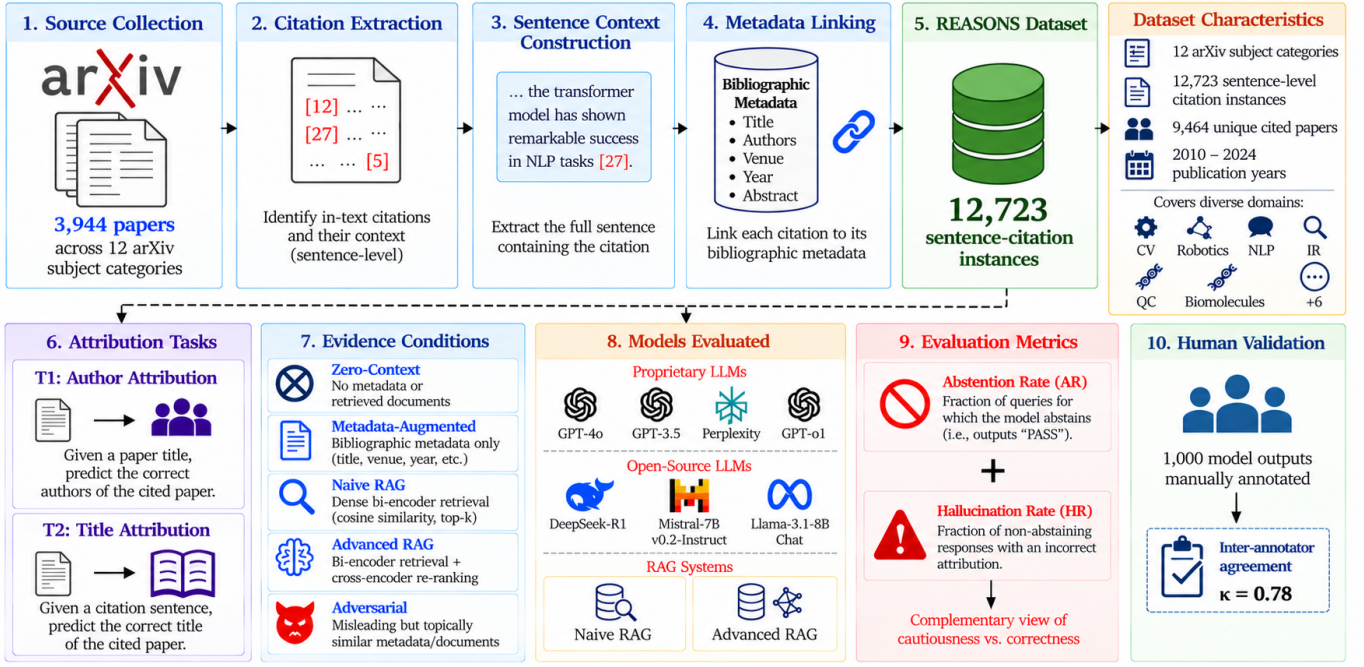


Fig. 1: Overview of the REASONS benchmark construction and evaluation framework.

II. BACKGROUND AND RELATED WORK

Attribution optimization in LLMs has focused primarily on improving citation correctness through better retrieval or training. RETRO [17] retrieves text chunks during pretraining to ground generation in external knowledge, while retrieval-augmented generation (RAG) [9] combines dense retrieval with sequence-to-sequence models for knowledge-intensive tasks. Source-aware training methods [11, 18] teach models to associate generated content with source identifiers, and self-supervised alignment techniques [10] refine citation placement through contrastive learning. Attributed QA [12] evaluates whether model outputs are supported by retrieved passages. While these approaches demonstrate that retrieval and training can improve citation accuracy, they evaluate whether systems *can* cite correctly when evidence exists, not whether systems *should* abstain when evidence is insufficient or misleading. This leaves unanswered the behavioral question raised in our introduction: when generating citations, do LLMs know when they don’t know, or do they confidently fabricate sources despite uncertainty?

Evaluation frameworks for attribution have largely focused on verifying whether generated content is supported by retrieved evidence. Benchmarks such as Attributed Question Answering (AQA) [12], ALCE [13], and related knowledge-grounding and fact-verification benchmarks [14, 19] assess whether generated responses or claims are supported by relevant evidence. Similarly, citation-intent datasets and large scholarly corpora [20, 21] support citation analysis, scholarly document understanding, and modeling of citation relationships. Although these benchmarks provide important measures of citation accuracy and retrieval effectiveness, they

primarily evaluate whether systems *can* cite correctly when evidence is present. Existing evaluation metrics are also based on precision, recall, or support-based scoring, which cannot distinguish systems that achieve high precision by abstaining from uncertain cases from those that respond to all queries while producing confident attribution errors. Consequently, these frameworks do not capture the fundamental behavioral tradeoff between abstention and hallucination, nor do they examine how models behave when evidence is incomplete, ambiguous, or misleading—conditions that frequently arise in real-world knowledge-seeking settings. Table I situates REASONS relative to existing datasets and shows that it is the first benchmark designed to systematically measure abstention–hallucination tradeoffs in the generation of citations.

Recent work on adversarial retrieval [22] reveals that LLMs exhibit attribution bias when retrieved passages are topically relevant but do not support specific claims, exposing a critical gap between retrieval quality (finding documents with high semantic similarity) and attribution safety (knowing when retrieved evidence is insufficient for the specific claim being made). Unlike search engines that present ranked results for user inspection, generative RAG pipelines convert retrieved evidence directly into assertions, raising the possibility that surface similarity might trigger confident citation regardless of claim-level support. This suggests that retrieval might make systems more confident without making them more correct, directly relevant to our question of whether adding evidence causes systems to shift from saying “*I don’t know*” to asserting wrong answers with high confidence.

Domain adaptation studies [23] show that task performance varies across domains, with general-purpose LLMs exhibit-

Dataset	Granularity	Domains	Abstention	Task
S2ORC [21]	Document	General	No	Scholarly Corpus
UnarXive [25]	Section	CS	No	Citation Analysis
RefSum [26]	Paragraph	CS	No	Summarization
SciCite [20]	Sentence	Comp. Ling.	No	Citation Intent Classification
REASONS (ours)	Sentence	12 research areas	Yes	Attribution

TABLE I: Comparison of REASONS with existing citation datasets.

Domain	Paper Count	Papers in IEEE format	Citation Count
Computer Vision (CV)	5488	1028	3437
Robotics	3656	292	776
Graphics	1796	384	1417
Information Retrieval (IR)	1741	564	1654
Artificial Intelligence (AI)	1697	531	2021
Natural Language Processing (NLP)	1526	293	1092
Cryptography	1084	371	1106
Neurons and Cognition (NNC)	892	111	326
Human Computer Interaction (HCI)	761	112	229
Databases	723	115	182
Quantum Computing (QC)	421	126	456
Biomolecules	119	17	27
Total	19904	3944	12723

TABLE II: Our benchmark dataset, REASONS, includes papers and sentences from 12 arXiv subject categories

ing stronger capabilities in well-represented areas such as Computer Vision than in specialized fields such as Quantum Computing, due to differences in pretraining corpus composition. However, whether attribution safety, the joint behavior of abstention and hallucination, degrades systematically with domain complexity or varies stochastically with model capacity remains unknown. This gap is critical for deployment: systems may perform well on aggregate benchmarks dominated by popular domains, yet exhibit unsafe behavior on specialized queries where epistemic caution is most critical. This concern is particularly relevant in high-stakes domains, where reliability and safety are central requirements for trustworthy AI deployment [24]. Existing work leaves our behavioral questions unanswered. Sentence-level granularity is not perfect: some claims require multi-sentence discourse, tables/figures, or broader context to verify. Our evaluation settings explicitly vary the amount of context available (e.g., minimal vs. metadata/retrieval conditions), allowing us to study how systems trade off abstention and hallucination as evidence becomes more complete. Sentence-level evaluation provides a controlled lens for observing attribution behavior under systematically manipulated evidence.

III. DATASET CONSTRUCTION AND BENCHMARK

As a central contribution, REASONS (REtrieval and Automated citationS Of scieNtific Sentences) is a sentence-level citation attribution benchmark comprising 12,723 citation instances extracted from 3,944 arXiv papers spanning 12 subject categories. Rather than artificially balancing domain representation, the dataset preserves the natural long-tail distribution (Table II), enabling systematic analysis of how attribution reliability varies between well-represented and highly specialized fields. This design allows us to disentangle whether

REASONS dataset example (Computer Vision)	
Link	http://arxiv.org/abs/2012.05435v2
Paper title	<i>Optimization-Inspired Learning with Architecture Augmentations and Control Mechanisms for Low-Level Vision</i>
Sentence ID	32
Sentence	We adopt the ℓ_1 -norm. For GM, we establish a residual network with seven convolutional layers and six ReLU blocks, each placed after a convolutional layer. The DM is constructed as a standard CNN-based classifier.
Citation text	C. Ledig et al., “Photo-Realistic Single Image Super-Resolution using a Generative Adversarial Network,” CVPR 2017.
Cited paper id	arXiv:1609.04802
Cited paper title	<i>Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network</i>
Abstract (excerpt)	Despite breakthroughs in accuracy and speed for single-image super-resolution using deep convolutional neural networks, ...
Authors	Christian Ledig, Lucas Theis, Ferenc Huszar, Jose Caballero, Andrew Cunningham, Alejandro Acosta, Andrew Aitken, Alykhan Tejani, Johannes Totz, Zehan Wang, Wenzhe Shi

TABLE III: Example instance from REASONS showing citation context and linked metadata.

attribution failures arise from domain specialization or simply reflect differences in data availability. Each instance contains a structured attribution datapoint linking the citing sentence to verified citation metadata, including the cited paper’s title, authors, abstract, and identifier (Table III). In addition to domain-level diversity, REASONS includes cross-domain citation markers that capture interdisciplinary references, providing controlled test cases in which topical similarity between citing and cited papers is reduced. These features enable fine-grained evaluation of attribution behavior under varying levels of semantic overlap and evidence availability. Link for dataset <https://zenodo.org/records/18264033>.

Metadata enrichment and structure: Each instance pairs a citing sentence with verified metadata of the referenced paper, including title, abstract, author list, venue, and publication year. This metadata is verified by exact matching of in-text citation strings to retrieved records. The source explicitly cited by the original authors is treated as ground truth; thus, REASONS measures fidelity to the observed citation rather than whether alternative sources could also support the claim. The dataset also records the source paper identifier and publication date, enabling analyses of temporal factors such as model knowledge cutoffs.

a) Tasks: We evaluate attribution behavior through two complementary tasks that probe different aspects of citation knowledge. *Task T1 (Direct Attribution)* provides the model with a paper title and asks it to return the author list or Abstain if uncertain. This tests whether models can ground metadata when the source identity is explicitly given, isolating their ability to recall or verify authorship information. *Task T2 (Indirect Attribution)* provides a citing sentence with an inline citation placeholder and asks the model to identify the cited paper’s title or abstain. This simulates deployed settings in which models must infer which source supports a claim

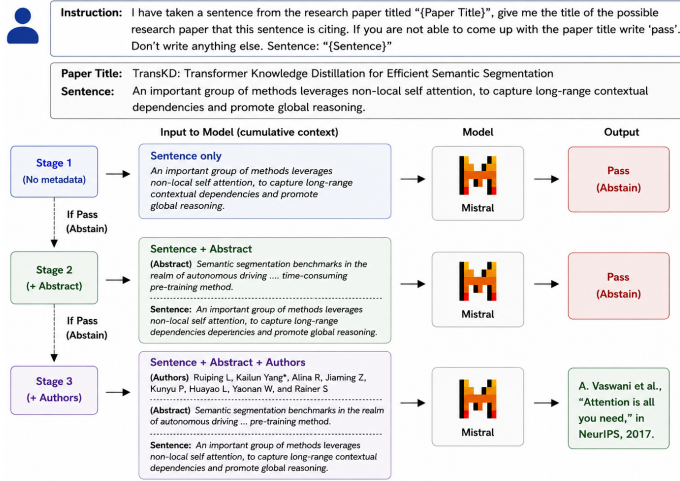


Fig. 2: CMP prompting as a staged attribution decision process. Under zero-shot indirect prompting (Task T2), the model abstains when asked to identify the cited paper from a sentence alone. As additional metadata is progressively revealed—first the abstract, then author information—the model transitions from abstention to confident attribution.

without author-provided citations, thereby testing attribution behavior when source identity must be determined. These tasks separate memorization of paper-author mappings (T1) from claim-to-source attribution reasoning (T2) (prompts in the supplementary).

b) Prompting Strategies: We examine three prompting strategies. Let x denote the query input (paper title for T1, citing sentence for T2), m denote citation metadata (abstract, authors), and $f(\cdot)$ the model’s generation function producing either an attribution $\hat{c}$ or the abstention token `Abstain` (PASS). In *zero-context prompting*, models receive only the query ($\hat{c} = f(x)$) and must either rely on parametric knowledge or abstain, thereby exposing baseline attribution behavior without retrieval. In *metadata-augmented prompting*, models receive verified citation metadata alongside the query ($\hat{c} = f(x, m)$), revealing how structured evidence affects the abstention-hallucination tradeoff. In *Cascaded metadata-augmented prompting (CMP)*, the model first attempts T2 with no context; if it abstains or produces an error, we then provide metadata and re-query, enabling analysis of how models revise attribution decisions as evidence accumulates. This staged disclosure tests whether additional context shifts systems from cautious abstention to confident error. Figure 2 illustrates CMP as a staged attribution decision process. CMP is not proposed as a training or calibration method but as a diagnostic tool: by staging evidence disclosure, it isolates whether a model’s abstention behavior is driven by genuine uncertainty or simply by the absence of context. The key finding is that models with poor uncertainty calibration cannot be rescued by better prompting alone, motivating the need for calibration-aware training as future work.

c) Retrieval Strategies: Let $\mathcal{D}$ denote the retrieval corpus from REASONS. We evaluate two retrieval conditions that differ in how supporting documents are selected and provided to the model. x is the query input, $f(\cdot)$ the model’s generation function, and $\hat{c}$ either an attribution or the abstention token

ABSTAIN.

Naive retrieval augmentation. A bi-encoder maps queries and documents into a shared embedding space. Documents are retrieved based on cosine similarity: $d_i \in \text{Top-}K_{d \in \mathcal{D}} \langle E(x), E(d) \rangle$, where $E(\cdot)$ denotes the embedding function and $\langle \cdot, \cdot \rangle$ denotes dot-product similarity. The top- k retrieved documents are concatenated with the query: $\hat{c} = f(x, d_1..d_k)$. This represents standard dense retrieval without re-ranking.

Advanced retrieval augmentation. The bi-encoder first retrieves a larger candidate set, which is then re-ranked using a cross-encoder that jointly scores each query-document pair. Documents are reordered according to $\pi(d | x) = \text{rank}(g(x, d))$, where $g(\cdot, \cdot)$ is the cross-encoder scoring function. The top-ranked documents from this reordered set are supplied to the generator: $\hat{c} = f(x, d_{\pi(1)}..d_{\pi(k)})$. The cross-encoder is fine-tuned on REASONS using listwise context loss to optimize re-ranking. All retrieval conditions index the same corpus, $\mathcal{D}$, to ensure a controlled comparison.

d) Models: We evaluate seven LLMs: GPT-o1 (G1), GPT-4o (G2), GPT-3.5-Turbo (G3), pplx-7b-chat (P), DeepSeek-R1 (D), Mistral-7B-v0.2-Instruct (M), and Llama-3.1-8B-Chat (L). All models are evaluated on T1 and T2 under zero-context, metadata-augmented, and CMP conditions, as applicable. For retrieval-augmented evaluation, we use Mistral and Llama as generators with Naïve RAG (RM, RL) and Advanced RAG (AM, AL). Naïve RAG retrieves the top- $k = 2$ documents, while Advanced RAG retrieves $k = 40$ candidates and re-ranks them using a cross-encoder. Complete model and retrieval configurations are provided in the Supplementary Material.

Citation-aware systems: We additionally evaluate GraphRAG [8], HLM-Cite [27], and LitFM [28] on T2. We provide the citing sentence as input and compare the top-ranked citation against ground truth; incorrect predictions are counted as hallucinations and null outputs as abstentions. These systems are not evaluated on T1 because they perform claim-to-citation mapping rather than metadata lookup. Results are reported separately in Table V; implementation details are provided in the Supplementary Material, with retrieval components summarized in Supplementary Table VII.

e) Evaluation Metrics: We evaluate attribution behavior using metrics that distinguish abstention, incorrect attribution (hallucination), and partial surface overlap, rather than reducing performance to accuracy alone.

Abstention Rate (AR). AR measures abstention frequency:

$$\text{AR} = \frac{1}{|\mathcal{Q}|} \sum_{i \in \mathcal{Q}} \mathbb{I}[\hat{c}_i = \text{PASS}].$$

where $\mathcal{Q}$ denotes the set of evaluation queries. AR reflects epistemic caution, not correctness. Abstention lacks instance-level ground truth—whether models should abstain depends on evidence availability. We validate AR through aggregate trends: it decreases with verified metadata but increases under adversarial perturbations (Supplementary Table VIII), demon-

strating sensitivity to evidence quality rather than prompt compliance.

Hallucination Rate (HR). HR measures incorrect attribution frequency conditional on responding. Here, hallucination denotes any non-abstaining attribution that does not match the benchmark ground truth, including both fabricated citations and incorrect attribution to real sources. Let $\mathcal{Q}_{\neg\text{PASS}} = \{i \in \mathcal{Q} \mid \hat{c}_i \neq c_i\}$:

$$\text{HR} = \frac{1}{|\mathcal{Q}_{\neg\text{PASS}}|} \sum_{i \in \mathcal{Q}_{\neg\text{PASS}}} \mathbb{I}[\hat{c}_i \neq c_i].$$

Citation correctness is determined through exact string matching after standard text normalization. For title attribution (T2), the generated title must match the ground truth exactly. For author attribution (T1), we require the generated list to contain precisely the same authors as ground truth. Author matching is order-invariant and handles standard name variations (e.g., A. Smith $\equiv$ Andrew Smith). We treat partial matches as incorrect; incomplete author lists or truncated titles count as attribution failures. This strict criterion ensures HR measures identity-level precision rather than approximate similarity.

Joint interpretation. AR and HR must be interpreted together to characterize epistemic safety. A system with high AR and low HR is cautious, it frequently declines to answer but rarely hallucinates when it does respond. Conversely, a system with low AR and high HR is overconfident; it answers most queries but produces many incorrect attributions. To illustrate the extremes: a model that always abstains achieves AR = 100% with zero hallucinations but provides no utility, while a random citation from our corpus yields HR $\approx$ 99.99% (only 1 in 12,723 citations would be correct by chance). Effective attribution systems must balance informativeness (low AR, high frequency of answers) with reliability (low HR, high accuracy). BLEU and F1 supplement HR to distinguish near-miss from spurious errors; correctness is validated through human evaluation on 1,000 stratified outputs.

Evaluation Process: We report performance aggregated across the 12 REASONS domains. Domain-level variability is reported using standard deviations in the Supplementary Material. Retrieval indices and re-ranking components remain fixed across all generator models, ensuring that observed differences reflect attribution behavior rather than variations in retrieval quality. Complete generation parameters, prompt templates, and human validation protocols are provided in the Supplementary Material.

IV. RESULTS

Figure 3 reports AR (refusal rate) and HR (error rate when answering), with error bars showing variability across 12 domains, revealing model caution versus overconfidence.

Baseline attribution behavior under minimal context: Under zero-context prompting, attribution is unreliable for both tasks, but failures differ by task type. For T1 (direct attribution), proprietary models achieve moderate AR ($\approx$ 35–40%) yet still exhibit substantial HR ($>$ 30%) among non-abstaining outputs (Figure 3a). Smaller or task-specific

Category	Domain	# Papers	HR (Zero-Context)	HR (Metadata)
Well-represented	CV	5488	71.3%	12.8%
	IR	1741	68.9%	14.2%
Specialized	QC	421	94.6%	31.7%
	Bio	119	96.1%	28.9%

TABLE IV: Hallucination Rate (HR) stratified by domain representation. Specialized, low-resource domains exhibit substantially higher epistemic risk than well-represented domains, even when metadata grounding is provided.

systems show extremely high HR (often $>$ 90%), indicating strong “authority-assertion” tendencies even when evidence is absent. T2 (indirect attribution) is consistently harder: HR increases sharply across models (often $>$ 70%), and model strategies diverge. GPT-4-class models respond by abstaining heavily (AR approaching 90%), while RAG/task-specific systems continue answering despite high error (Figure 3c). Zero-context diagnoses whether a model defaults to calibrated caution (high AR, low HR) or overconfident fabrication (low AR, high HR).

Effect of metadata on attribution behavior: Adding verified metadata drastically reduces HR for T1, and AR collapses; models answer because the evidence is explicit (Figure 3b). Metadata reduces error, but can also shift the failure mode. For T2, HR drops but remains nontrivial, while AR often approaches zero for many models despite residual mistakes. T1 becomes almost “copy-from-evidence.” Authorship is explicitly present in metadata, so the task is closer to extraction than inference. T2 still requires alignment and selection. Even with metadata, models must connect a cited claim to the correct paper/title among plausible candidates; this invites surface similarity bias (choosing what “looks right” rather than what is correct). Metadata can create evidence-induced confidence: models stop abstaining because something looks grounded, but the grounding may be misapplied (wrong paper/title is selected), leaving HR at 15–20% for some models.

RAG models: RAG variants reduce HR relative to zero-context baselines and improve F1/BLEU, but they often show near-zero AR even when HR remains high (Table V). Naive RAG is especially prone to confident errors; advanced RAG reduces HR but does not restore abstention. Retrieval injects text that looks like evidence, which increases the model’s tendency to answer. Therefore, retrieval strength is not a proxy for reliability; AR–HR jointly identify whether RAG is helping correctness or merely suppressing abstention.

CMP reduces HR and improves attribution quality relative to zero-context, but model families respond differently: GPT-4-class models retain higher AR, while RAG-based systems still suppress AR and keep asserting attributions (Figure 3d). Staging prevents models from committing prematurely by forcing incremental reasoning. If a model always answers when given context, adding more stages does not help it abstain. CMP works best for models that already know when to say “I don’t know.” For overconfident models, it reduces errors somewhat but does not fix the abstention problem.

Domain effects across tasks (Specialized vs. Well-represented): Specialized domains (QC, Biomolecules) show

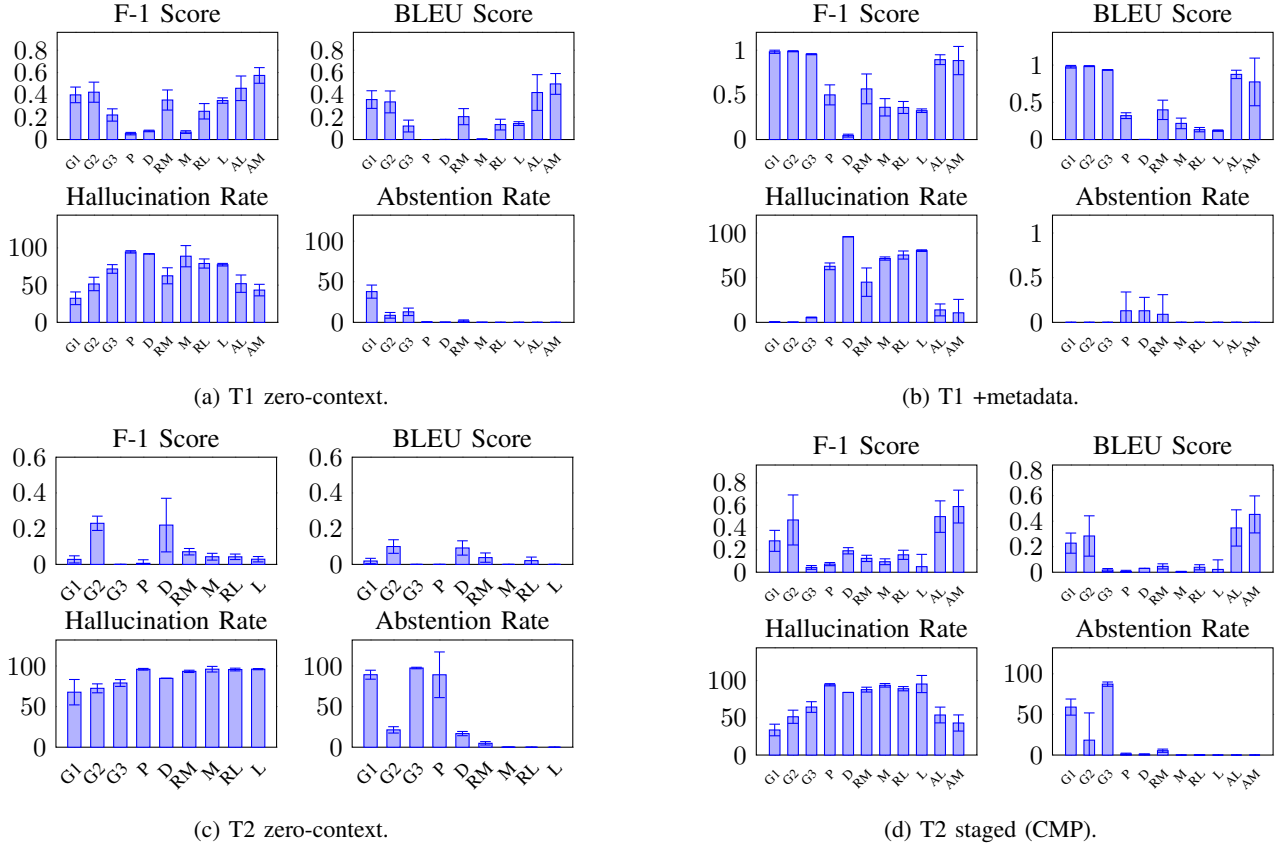


Fig. 3: Attribution behavior (AR/HR) across models and tasks, averaged over 12 REASONS domains. RAG-based systems (RM, RL, AL, AM) suppress abstention even when hallucination remains high, while G1 and G2 retain higher AR at the cost of responsiveness. Other models (G3, M, L, D, P) tend toward overconfident hallucination, revealing distinct epistemic risk profiles across model and retrieval configurations. Error bars show variability across the 12 domains.

Method	AR (%)	HR (%)	F1	BLEU
Naive RAG	5.03	87.56	0.14	0.09
Advanced RAG	0.00	65.41	0.58	0.32
GraphRAG	2.14	68.33	0.52	0.38
HLM-Cite	12.45	71.22	0.41	0.31
LitFM	8.91	74.16	0.48	0.35

TABLE V: Citation-aware systems trade abstention for responsiveness. GraphRAG exhibits near-zero AR, while HLM-Cite retains limited abstention. F1 and BLEU reveal that even lower-HR systems produce only partial matches, indicating that attribution errors are systematic rather than incidental.

consistently higher HR than well-represented ones (CV, IR), even with metadata (Table IV). Subsampling to equalize domain size preserves the gap, indicating the effect is not just data volume. Specialized fields use dense jargon, uncommon venues, and long-tail author/title distributions, conditions where “plausible guessing” is easier than correct identification. Retrieval is also noisier: similar terms or nearby topics produce highly plausible but wrong candidates. Domain specialization is a structural risk factor: even good prompting and retrieval can leave substantial residual HR, so deployment in specialized areas needs stronger verification than “more context.”

Summary: We observe three recurring behaviors: (1) cautious abstention (high AR, low HR); (2) confident hallucination (low AR, high HR); and (3) metadata-induced confidence, where

added context suppresses abstention without eliminating error. These regimes emerge from interactions among task type, evidence availability, retrieval design, and domain structure.

Human validation of HR Metric: To validate that HR accurately captures attribution behavior, we conduct a human evaluation on a stratified sample of **1,000** model outputs. These instances were stratified across three dimensions to ensure representativeness: (a) *Model coverage*: 200 instances each from GPT-o1, GPT-4o, Advanced RAG (Mistral), Perplexity AI (7B), and DeepSeek; (b) *Domain diversity*: Proportional sampling across well-represented domains (CV, IR) and specialized domains (QC, Biomolecules); (c) *Behavioral distribution*: Deliberate oversampling of abstention cases yields 500 non-pass instances (50%) and 500 pass instances (50%), enabling robust analysis of both attribution correctness and abstention appropriateness. This balanced distribution contrasts with the natural model behavior (which exhibits >90% non-pass rates) and provides sufficient power to analyze abstention decisions, a critical but underrepresented behavior in typical model outputs.

Annotation protocol. Three domain experts (PhD students with >3 years of research experience) independently evaluate each instance. Annotators receive: (1) the input query (paper

Model	n (non-pass)	Auto HR (%)	Human Incorrect (%)	r
G1	98	17.99	19.39	0.91
G2	97	17.59	18.39	0.85
AM	102	75.41	73.53	0.88
P	101	86.78	84.16	0.87
D	102	71.34	69.61	0.90
Total	500	–	–	0.89

TABLE VI: Human evaluation validates automated HR measurement. HR is computed only over non-pass instances and compared against human judgments of attribution correctness on the same subset. Strong correlation ($r = 0.89$, $p < 0.001$) confirms that lexical-level metrics capture human-perceived errors with minimal confounding from acceptable paraphrases.

title for T1, cited sentence for T2); (2) the model output (attribution or PASS); (3) ground-truth citation metadata (title, authors, abstract); (4) evidence condition (zero-context, metadata-augmented, or CMP).

Annotators apply a two-stage judgment process depending on model behavior: *Stage 1. For non-pass outputs (n=500):* Rate attribution correctness: (a) Correct: Model output matches ground truth semantically, allowing minor formatting variations (e.g., “Reliable Attribution in Scientific Language Models” vs. “A Trustworthy Attribution for Scholarly Language Models”); (b) Partially Correct: Model identifies the correct paper but with incomplete information (e.g., missing co-authors, abbreviated title); (c) Incorrect: Model output does not match ground truth (hallucination).

Stage 2—Pass outputs (n=500): Annotators judged whether each abstention was appropriate given the available evidence. An abstention was marked appropriate if the model correctly declined when evidence was insufficient (e.g., zero-context with an ambiguous query), and unnecessary if the model declined despite having sufficient evidence (e.g., complete metadata provided).

Inter-annotator agreement. Fleiss’ κ was computed separately for attribution correctness ($\kappa = 0.81$) and abstention appropriateness ($\kappa = 0.74$), yielding an overall $\kappa = 0.78$, indicating substantial agreement across all judgment types. Disagreements were resolved by majority vote. Among the 500 non-pass instances, 330 (66.0%) were judged factually incorrect. Among the 500 pass instances, 271 (54.2%) abstentions were judged appropriate and 229 (45.8%) unnecessary.

Validation of automated metrics. We validate automated HR by comparing it to human-judged error rates on the same outputs (Table VI), confirming strong alignment despite minor paraphrase variation. The strong correlation ($r = 0.89$, $p < 0.001$) between automated HR and human-judged error rates demonstrates that: 1) Lexical-level metrics effectively capture attribution errors without being confounded by acceptable paraphrases. 2) When models hallucinate, they typically produce factually different content (wrong paper, wrong authors), not mere rewordings. 3) Small divergences (e.g., G1: 17.99% automated vs. 19.39% human) reflect cases where human experts identified subtle errors (author name typos, venue misattributions) that lexical metrics missed.

Only 5.2% of non-pass outputs corresponded to acceptable

paraphrases, indicating that most detected errors were factual attribution failures rather than lexical variations. *Additional analyses of paraphrase and abstention behavior are provided in the Supplementary Material.*

V. ETHICAL CONSIDERATIONS

The dataset consists of publicly available arXiv papers; we follow arXiv usage policies, exclude restricted content, and use no private or personal data. This is an observational study of system behavior, and we do not release fine-tuned generator checkpoints trained on restricted material. We evaluate two tasks: **T1** (direct attribution), which tests whether systems assert or withhold authority cues, and **T2** (indirect attribution), which tests whether systems link claims to sources. Across both tasks, we characterize behavior via the joint dynamics of **AR** and **HR**; unless stated otherwise, results are averaged over all 12 domains. Because REASONS is derived from arXiv, some benchmark instances may overlap with LLM pretraining corpora. T1 may partly reflect memorized title–author associations, whereas T2 requires claim-to-source attribution and is therefore less directly dependent on metadata recall.

REASONS evaluates attribution fidelity to original author citation choices, not attribution correctness in an absolute sense. Authors may miscite or overcite, meaning a model that identifies a more appropriate citation than the original may still be penalized. Author citation choices provide scalable ground truth for this benchmark, and the same evaluation criterion is applied consistently across models. The human annotations are not released because of institutional review requirements; however, we release the annotation rubric, protocol, examples, and aggregation procedure to facilitate reproduction.

VI. CONCLUSION

We introduced REASONS, a sentence-level benchmark for scientific source attribution across 12 arXiv subject categories, and used it to systematically evaluate how LLMs attribute paper *titles* and *authors* under varying evidence conditions. Our experiments reveal a consistent *abstention–hallucination* tradeoff: models that answer more readily often reduce apparent abstention but may increase confident misattributions, especially when contextual cues resemble authoritative metadata. RAG and enriched citation context typically improve attribution fidelity, yet can also suppress abstention even when the retrieved evidence is incomplete or misleading, creating a different form of reliability risk. Our results suggest that evaluating scientific attribution systems requires metrics beyond accuracy alone, explicitly accounting for when models should abstain. Our findings are limited to the 12 arXiv subject categories studied and should not be assumed to generalize to other scientific disciplines. Future work includes improving evidence-grounding under noisy retrieval, designing calibration methods that preserve appropriate abstention, and extending the benchmark to broader citation styles and full-document attribution settings.

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SUPPLEMENTARY MATERIAL

The REASONS dataset is publicly available on Zenodo (<https://zenodo.org/records/18264033>). The supplementary material provides prompt templates, domain-level evaluation results, model outputs, configuration files, retrieval and generation settings, and additional documentation required for reproduction.

A. Prompt Templates and Model Specifications

Task T1: Author Attribution

Given the following paper title, identify the complete list of authors. If you are uncertain, output PASS.

Input: Paper title

Output: Author list or PASS

Task T2: Title Attribution

Given the following citation sentence, identify the cited paper title. If you are uncertain, output PASS.

Input: Citation sentence

Output: Paper title or PASS

Metadata-Augmented Prompt (T2)

Given the following citation sentence and metadata, identify the cited paper title.

If you are uncertain, output PASS.

Sentence: *citation sentence*

Abstract: *paper abstract*

Authors: *author list*

Output: Paper title or PASS

CMP Prompt (T2)

Identify the cited paper title.

If uncertain, output PASS.

Stage 1 Input: Sentence only

Stage 2 Input: Sentence + Abstract

Stage 3 Input: Sentence + Abstract + Authors

Output: Paper title or PASS

Model Specifications

- **GPT-o1 (G1):** OpenAI API with default decoding parameters.
- **GPT-4o (G2):** OpenAI API, temperature = 1.0, top-p = 1.0, max tokens = 256.
- **GPT-3.5-Turbo (G3):** OpenAI API, temperature = 1.0, top-p = 1.0, max tokens = 256.
- **pp1x-7b-chat (P):** evaluated through the Perplexity API.
- **DeepSeek-R1 (D):** greedy decoding (do_sample=False).
- **Mistral-7B-v0.2-Instruct (M):** temperature = 1.0, top-p = 0.95, max new tokens = 300.
- **Llama-3.1-8B-Chat (L):** temperature = 1.0, top-p = 0.95, max new tokens = 300.

Experiments were conducted on NVIDIA A100 and L40S GPUs. Resources will be released upon publication. For zero-context, metadata-augmented, and CMP experiments, API-based models (GPT-4o, GPT-3.5, Perplexity) were evaluated using their standard sampling configuration (tempera-

Component	Model
Embedding Model	BAAI/bge-small-en-v1.5
Cross Encoder	ms-marco-MiniLM-L-12-v2
Llama Tokenizer	NousResearch/Llama-3.1-8b-chat-hf
Mistral Tokenizer	mistralai/Mistral-7B-v0.2

TABLE VII: Retrieval and tokenization components.

ture=1.0, top-p=1.0). For RAG-based experiments using Mistral and Llama models, generation employed stochastic decoding (temperature=1.0, top-p=0.95, max_new_tokens=300). For DeepSeek adversarial and retrieval experiments, greedy decoding (do_sample=False) was used to ensure reproducible attribution behavior under controlled retrieval conditions.

All implementation details, including generation parameters (temperature, max tokens, top-p), embedding models, retrieval hyperparameters, and top-*k* values, are provided in Suppl. Table VII. The cost associated with this research includes expenses for utilizing OpenAI API, totaling \$640.37. Additionally, the use of Perplexity API incurred costs amounting to \$259.39. Furthermore, GPU resources, we used Replicate¹ API for our experiments, amounted to \$466.22. For dataset creation, we used Oxylab for \$249 for a month. In total, the expenses for APIs were \$1614.98.

Quality Control and Challenging Instances: We validate citation instances through exact matching between in-text citation strings and verified bibliographic metadata, removing unmatched or malformed entries. However, we intentionally retain citation contexts that are fragmented, numeric-heavy, or difficult to interpret in isolation. Such cases naturally occur in scientific writing and represent realistic attribution challenges. Preserving these difficult examples enables a more rigorous evaluation of abstention and hallucination behavior, rather than restricting the benchmark to only clean, self-contained citation contexts.

B. Adversarial attribution setting:

To examine how attribution behavior changes when credibility cues are misleading, we introduce an adversarial evaluation setting. In this setting, the attribution task remains unchanged, but the metadata provided to the model is deliberately perturbed to be plausible yet incorrect. This allows us to test whether systems rely on surface-level similarity rather than contextual alignment when justifying sources. Let x denote the query input (paper title for Task T1 or cited sentence for Task T2), and let m denote the correct citation metadata. Under standard conditions, the model produces an attribution $\hat{c} = f(x, m)$. For adversarial evaluation, we replace m with perturbed metadata $\tilde{m}$ such that $\tilde{m} \neq m$ but remains highly similar: $\hat{c}^{adv} = f(x, \tilde{m})$, where $\text{sim}(m, \tilde{m}) \geq \tau$. Here, $\text{sim}(\cdot, \cdot)$ denotes a similarity function and $\tau \geq 0.85$ is a threshold chosen to ensure substitutions are realistic and difficult to distinguish based on surface cues alone. For Task T1, adversarial inputs are constructed by replacing the correct paper title with a similar but incorrect title. For Task T2, adversarial inputs are created by substituting either

¹<https://replicate.com/>

Group	AR (%)	HR (%)	BLEU
Adversarial Setting: Changing Paper Title (Task T2)			
GPT-o1 (reasoning)	96.23	17.99	0.6210
GPT-4o	31.45	83.66	0.0524
GPT-3.5	68.55	87.35	0.0389
D	89.31	61.31	0.0178
RAG(M)	3.14	86.78	0.0796
RAG(L)	5.03	87.56	0.0628
AdvRAG(L)	0.00	85.72	0.1322
AdvRAG(M)	0.00	75.41	0.1569
Adversarial Setting: Changing Paper Abstract (Task T2)			
GPT-o1 (reasoning)	95.60	38.49	0.4595
GPT-4o	32.70	86.22	0.0396
GPT-3.5	76.10	91.64	0.0034
DeepSeek	93.08	94.72	0.0000
RAG(M)	7.55	89.44	0.0520
RAG(L)	2.52	90.16	0.0445
AdvRAG(L)	0.00	39.67	0.4101
AdvRAG(M)	0.00	39.57	0.4904

TABLE VIII: Attribution behavior under adversarial metadata in Task T2. Either the cited paper title or abstract is replaced with a highly similar but incorrect alternative. Adversarial metadata substantially increases hallucination while suppressing abstention, revealing a tendency to assert sources under misleading credibility cues.

the cited paper title or abstract with a highly similar alternative. No additional training or fine-tuning is performed in this setting. This isolates attribution behavior under misleading evidence and allows us to observe whether systems abstain or confidently assert incorrect attribution.

Attribution behavior under adversarial metadata: When titles/abstracts are replaced with highly similar but incorrect alternatives, HR increases sharply and AR often decreases (Table VIII). Some proprietary models increase AR, but retrieval-driven systems commonly continue to answer. Models overweight semantic proximity and “credibility signals” (paper-like text), treating them as confirmation. RAG pipelines can inadvertently validate the wrong candidate by retrieving a coherent-looking neighbor document. This is not just a worst-case attack; it demonstrates a common real-world failure: near-duplicate literature and ambiguous citations can trigger confident misattribution.

C. Human Evaluation

Analysis of paraphrase cases. Among the 500 non-pass instances evaluated, only 26 (5.2%) involved semantically correct paraphrases that automated metrics incorrectly penalized (e.g., “Reliable Attribution in Scientific Language Models” vs. “Trustworthy Attribution for Scholarly Language Models”, author name variations like “A. Vaswani” vs. “Ashish Vaswani”). Conversely, 330 instances (66.0%) were factually incorrect hallucinations that both automated metrics and human experts identified (wrong paper, wrong authors, fabricated titles). This 12.7:1 ratio (330:26) confirms that hallucination is predominantly a factual error problem, not a paraphrase-

detection problem, thereby validating our automated measurement approach. The remaining 144 instances (28.8%) were judged as “Partially Correct” by human annotators—cases where models identified the correct general paper but with incomplete or slightly inaccurate information (e.g., missing co-authors, abbreviated journal names, minor title variations that changed meaning). Our automated HR metric conservatively counts these as errors, accounting for the slight divergence between automated and human-judged error rates in Table VI. **Abstention appropriateness findings.** Across the 500 pass instances, human experts judged 271 abstentions (54.2%) as appropriate (the model correctly declined when evidence was insufficient, e.g., zero-context ambiguous queries) and 229 (45.8%) as unnecessary (the model could have answered given available evidence, e.g., when complete metadata was provided). This near-even split indicates that current LLMs miscalibrate abstention decisions, oscillating between over-caution when evidence suffices and answering when abstention is warranted. It also motivates evaluating AR and HR jointly rather than optimizing either in isolation: very low AR can yield confident but incorrect answers (epistemic risk), while very high AR can lead to unnecessary abstentions (reduced utility).